

Practical-5**Practical Title: Registration Form with Frontend Validation and User-Friendly Error Handling**

Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

Theory

1. Are correct input types and attributes used?
2. Are validation errors displayed near the relevant fields?
3. Is password strength checked?
4. Is the form accessible using keyboard and screen-reader-friendly labels?

Java script code:

```
function validateLogin() {
    let username = document.getElementById("username").value.trim();
    let password = document.getElementById("password").value.trim();

    if (username === "") {
        alert("Username is required");
        return false;
    }

    if (password.length < 6) {
        alert("Password must be at least 6 characters");
        return false;
    }

    alert("Login Successful!");
    return true;
}

function validateRegister() {
    let fullname = document.getElementById("fullname").value.trim();
    let studentid = document.getElementById("studentid").value.trim();
    let mobile = document.getElementById("mobile").value.trim();
    let email = document.getElementById("email").value.trim();
    let division = document.getElementById("division").value;
```

```
if (fullname === "") {
    alert("Full Name is required");
    return false;
}

if (studentid === "") {
    alert("Student ID is required");
    return false;
}

if (!/^[0-9]{10}$/.test(mobile)) {
    alert("Enter a valid 10-digit Mobile Number");
    return false;
}

if (!/^[^\\s@]+@[^\\s@]+\\.[^\\s@]+$/ .test(email)) {
    alert("Enter a valid Email Address");
    return false;
}

if (division === "") {
    alert("Division is required");
    return false;
}

alert("Registration Successful!");
return true;
}

function result(event) {

    if (event) {
        event.preventDefault();
    }

    const name = document.getElementById("name");
    const degree = document.getElementById("degree");
    const sem = document.getElementById("sem");
    const exam = document.getElementById("exam");
    const id = document.getElementById("ID");
```

```
const nameValue = name.value.trim();
const degreeValue = degree.value.trim();
const semValue = sem.value.trim();
const examValue = exam.value.trim();
const idValue = id.value.trim();

name.style.border = "1px solid #ccc";
degree.style.border = "1px solid #ccc";
sem.style.border = "1px solid #ccc";
exam.style.border = "1px solid #ccc";
id.style.border = "1px solid #ccc";

let isValid = true;

if (nameValue === "" || nameValue === "Select Institute") {
  name.style.border = "2px solid red";
  isValid = false;
}

if (degreeValue === "" || degreeValue === "Select Degree") {
  degree.style.border = "2px solid red";
  isValid = false;
}

if (semValue === "" || semValue === "Select Semester") {
  sem.style.border = "2px solid red";
  isValid = false;
}

if (examValue === "" || examValue === "Select Month") {
  exam.style.border = "2px solid red";
  isValid = false;
}

if (idValue === "") {
  id.style.border = "2px solid red";
  isValid = false;
}
```

```

    if (!isValid) {
        showNotification("Please fill in all required fields.");
        return false;
    }

    showNotification("Result retrieved successfully!");

    setTimeout(function() {
        window.location.href = "resultpage.html";
    }, 1000);

    return false;
}

/*----- Clear Form -----*/

function clearForm() {
    const form = document.querySelector("form");

    if (form) {
        form.reset();

        const inputs = form.querySelectorAll("input, select, textarea");

        inputs.forEach(function(input) {
            input.style.border = "1px solid #ccc";
        });

        showNotification("Form cleared successfully!");
    }
}

/*----- Contact Page -----*/

function contact() {

    const name = document.getElementById("name");
    const message = document.getElementById("message");

```

```

const nameValue = name.value.trim();
const messageValue = message.value.trim();

name.style.border = "1px solid #ccc";
message.style.border = "1px solid #ccc";

if (nameValue === "" || messageValue === "") {

    if (nameValue === "") {
        name.style.border = "2px solid red";
    }

    if (messageValue === "") {
        message.style.border = "2px solid red";
    }

    showNotification("Please enter both name and message.");
    return false;
}

showNotification("Message sent successfully!");

name.value = "";
message.value = "";

return false;
}

function showNotification(message) {
    const notification = document.getElementById("notification");
    const notificationMessage =
document.getElementById("notificationMessage");

    if (!notification || !notificationMessage) {
        return;
    }

    notificationMessage.textContent = message;

```

```

notification.classList.add("show");

setTimeout(function() {
  notification.classList.remove("show");
}, 3000);
}

function toggleMenu() {
  const navbar = document.getElementById("navbar");
  const menuButton = document.getElementById("menuButton");

  navbar.classList.toggle("active");

  if (navbar.classList.contains("active")) {
    menuButton.textContent = "×";
  } else {
    menuButton.textContent = "≡";
  }
}

function initTheme() {
  const savedTheme = localStorage.getItem("theme") || "light";
  const body = document.body;
  const themeButton = document.getElementById("themeBtn");

  if (savedTheme === "dark") {
    body.classList.add("dark-theme");
    if (themeButton) themeButton.textContent = "☀ Theme";
  } else {
    body.classList.remove("dark-theme");
    if (themeButton) themeButton.textContent = "🌙 Theme";
  }
}

function toggleTheme() {
  const body = document.body;
  const themeButton = document.getElementById("themeBtn");

  body.classList.toggle("dark-theme");

```

```

if (body.classList.contains("dark-theme")) {
    localStorage.setItem("theme", "dark");
    if (themeButton) themeButton.textContent = "☀ Theme";
} else {
    localStorage.setItem("theme", "light");
    if (themeButton) themeButton.textContent = "🌙 Theme";
}
}

window.addEventListener("DOMContentLoaded", function() {
    initTheme();
    const themeBtn = document.getElementById("themeBtn");
    if (themeBtn) {
        themeBtn.addEventListener("click", toggleTheme);
    }
});

```

Output:-

The screenshot shows a registration form with the following fields and values:

- Date of Birth: [Empty]
- Branch: [Empty]
- Semester: Select Semester
- Mobile Number: 9879129885
- Address: krishdesai301@gmail.com

Buttons: Register, Clear

Footer: © 2025 Student Portal. All Rights Reserved.

Theme

Student Portal

Student Registration Form

Full Name

krish desai

Enrollment ID

25dce019

Email

krishdesai301@gmail.com

Password

Enter password

Confirm Password

Confirm password

Gender

Male

Female

Other

Date of Birth

Branch

Computer Science

Semester

Mobile Number

9879129985

Address

krishdesai301@gmail.com

Register

Clear

Already have an account? [Login Here](#)

Back to Dashboard

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127.0.0.1:5500 says

Please fill in all fields correctly. Invalid fields are marked red.

OK

Email

kr

Phone Number

Enter your phone number

Subject

Maths

Message

Write your message

Send Message

Clear

College Contact Details

Address: Depstar College, near vadtal, Anand

Email: krishdesai301@gmail.com

Phone: +91 9876543210

Back to Dashboard

Student Portal

Theme

Contact Us

Full Name

Krish

Email

krishdesai301@gmail.com

Phone Number

Enter your phone number

Subject

Maths

Message

Write your message

Send Message

Clear

College Contact Details

Address: Depstar College, near vadtal, Anand

Email: krishdesai301@gmail.com

Phone: +91 9876543210

Back to Dashboard

B. GitHub Repository Link: <https://github.com/krish-301/Student-Hub-Portal>

C. Challenges Faced: The main challenge is implementing accurate **Regex and JavaScript validation** for all form fields.

Another challenge is providing **user-friendly errors, password strength checking, and accessibility**.

D. Learning Outcomes: tag. Students will learn to design accessible and user-friendly HTML registration forms.

They will implement JavaScript and Regular Expressions for accurate input validation.

They will learn password strength checking and effective error handling.

They will understand client-side validation for reliable data entry.

Faculty Remarks: